

Internet & API Fundamentals - Structured Notes

1. What is an API?

API (Application Programming Interface) is a bridge that allows two software systems to communicate.

Example flow: Client → API → Server → Database → Response.

API hides backend complexity and exposes only required data or actions.

2. Types of APIs: REST vs SOAP

REST:

- Uses HTTP methods (GET, POST, PUT, PATCH, DELETE)
- Mostly JSON format
- Lightweight and fast

SOAP:

- Uses XML only
- Strict protocol rules
- Heavy but highly structured

3. REST Principles

- Stateless: every request is independent
- Resource-based: URLs represent resources (/students/, /users/)
- Uniform interface: same URL pattern everywhere
- Client-server separation: frontend and backend independent
- Cacheable responses: improves performance

4. Request-Response Cycle

1. Client sends HTTP request
2. Server receives request
3. Business logic executes
4. Database accessed if needed
5. Response returned

Request contains:

- Method
- URL
- Headers
- Body (optional)

Response contains:

- Status code
- Headers

- Body

5. JSON Structure

JSON = JavaScript Object Notation

Object:

```
{  
  "name": "Rahul",  
  "age": 22  
}
```

Array:

```
{  
  "subjects": ["Math", "Physics"]  
}
```

Nested object:

```
{  
  "student": {  
    "name": "Rahul"  
  }  
}
```

6. REST vs SOAP Comparison

REST → Fast, simple, JSON, widely used

SOAP → Strict, XML, enterprise systems

7. DRF Connection

In Django REST Framework:

URL → View → Serializer → Model → Database

Database → Serializer → JSON Response

8. Common HTTP Methods

HTTP Method	Meaning
GET	Read data
POST	Create data
PUT	Full update
PATCH	Partial update
DELETE	Remove data